MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (90 minutes) December 3rd, 2011
Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1. In 2010 the known size of the ceiling tile markets in the US and China were

In the US - 2,000,000 Units ($\log_{10} = 6.30$)
In China - 1,000,000 Units ($\log_{10} = 6.00$)

and we want to make predictions about these market sizes in 2020

Let $X_1, X_2, \ldots\ldots, X_{10}$ be random variables representing the growth of the $\log_{10}$ ceiling tile market size in the US for the next 10 years with each X_i independent and normal with mean=0.1 and std-dev=0.2. For example the size of the $\log_{10}$US ceiling tile market in 2012 is $6.30 + X_1 + X_2$

Let $Y_1, Y_2, \ldots\ldots, Y_{10}$ be random variables representing the growth of the $\log_{10}$ ceiling tile market size in China for the next 10 years with each Y_i independent and Normal with mean=0.2 and std-dev=0.1.

A. What is the probability that the US market size will be smaller in 2020 than it was in 2010

B. Assume that for each year i, the correlation of X_i and Y_i is 0.5, but as before that growth between different years is independent. Derive a 95% prediction (i.e. confidence) interval for the ratio of the (China Market Size / US market size) in 2020.

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2. A bicycle manufacturer designs an experiment to compare two of their "off-the-road" bicycle models. Model 1 is the *Schwinn Frontier*. Model 2 is the *Schwinn High Timber*.

The manufacturer takes a sample of four bicycles of each model. A sample of 64 bicycle riders are assigned at random, 8 to each bicycle. Each bicycle rider will ride over a five-mile up-hill trail and the response variable *y*, is the time to complete the trip.

The manufacturer wants to see how gear-setting and tire pressure combinations affects the comparative performance of the bicycles. The experiment uses two gear settings (G1 and G2) and two tire pressures (P1 and P2). The 8 riders on a given bicycle are randomly assigned to the 4 combinations of levels of G and P: For 2 riders the bikes will be set at G1P1, for 2 at G1P2, for 2 at G2P1, and for 2 at G2P2.

For the ANOVA table corresponding to this experiment, write out the Sources of Variability. For each Source of Variability give degrees of freedom, Expected Mean Squares, and the appropriate denominator of the F statistic.

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3. We have the following model for the probability that a pine tree will survive

$P_{Survuval} \mid F = \dfrac{e^{\alpha+\beta F}}{1+e^{\alpha+\beta F}}$ where α and β are unknown parameters and F is the number of units of fertilizer given to the tree. The following MLE point estimates and standard deviations are obtained for α and β;

$\hat{\alpha} = 0.12 \pm 0.01, \hat{\beta} = 0.024 \pm 0.003$ with $Corr(\hat{\alpha}, \hat{\beta}) = -0.1$

A. Give a 95% confidence interval for the ratio of odds of survival for a tree given F=100 units over those of a tree given F=80 units of fertilizer

B. It costs $1000 for each tree that dies. Let C denote the cost per unit of fertilizer given to the tree. Find the MLE for the value of C at which F=80 units and F=100 have the same expected combined costs for fertilizer and dead trees.

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4. A medical researcher wanted to compare the effectiveness of two medicines A and B commonly prescribed for treatment of migraine headaches. As part of good experimental practice, there was a placebo, medication C that was used in the experiment. However the researcher was not particularly interested in comparing A or B to C.

The experiment took 100 individuals with chronic migraine headaches, and had each individual use all three medications, each medication for a one-week period. (The order of using the medications was randomized independently for each individual.) At the end of each week, each individual was asked "Did the medication you used during the past week help? (Yes or no). The following is a summary of the outcomes of the experiment for the 100 individuals:

Table 1:	Medicine A	Medicine B	Medicine C (placebo)
31 individuals said	No	No	No
10 individuals said	Yes	No	No
2 individuals said	No	Yes	No
14 individuals said	Yes	Yes	No
14 individuals said	No	No	Yes
4 individuals said	Yes	No	Yes
7 individuals said	No	Yes	Yes
18 individuals said	Yes	Yes	Yes

The researcher summarized the data for medicine A and B:

Table 2:	Medicine A	Medicine B
45 individuals said	No	No
14 individuals said	Yes	No
9 individuals said	No	Yes
32 individuals said	Yes	Yes

a) Based just on Table 2, is there evidence (statistically significant at 0.05 level) of a difference between A and B? Show your analysis, including formula for test statistic used, its computed values and conclusion of test. (Do not just say reject or accept, but explain in terms researcher can understand.)

b) From Table 1 we see that there are 43 individuals with chronic migraines who said the Placebo helped their headaches. Think about what this implies. Explain how you would analyze the data to see if there is a difference between the effectiveness of Medicine A and Medicine B.

c) Carry out the analysis you proposed in part (b) and explain your conclusions.